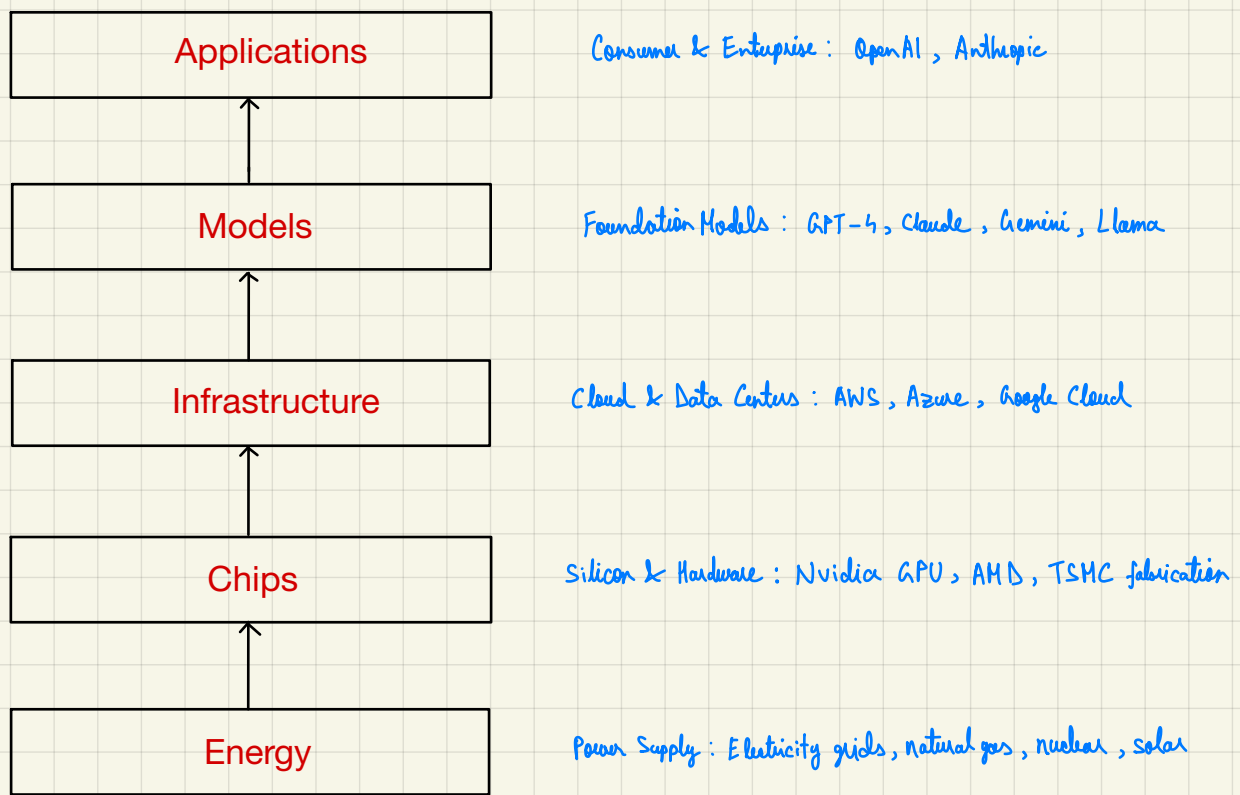
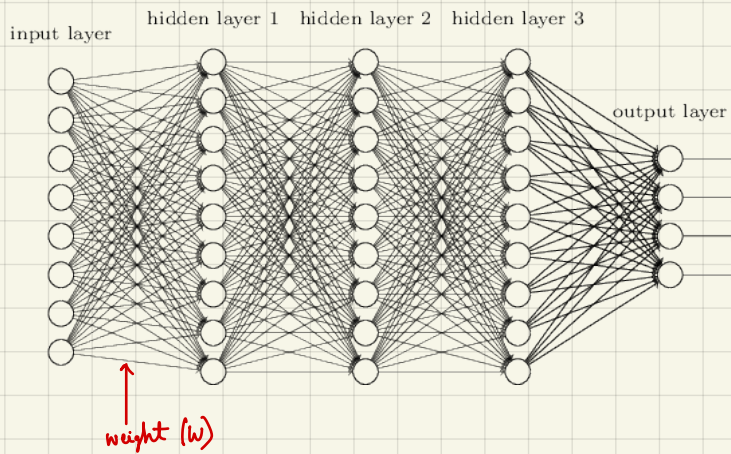




Computing model inspired by human brain consisting of interconnected layers of nodes/neurons that process data to identify patterns

Node - Thing that holds a number $\rightarrow$ Function



Neuron inside human brain either fire completely or none.

Contrast, nodes can do partial firing and the amount of firing is called Activation (a)

Activations should be inside a range $\Rightarrow$ Sigmoid function $\sigma(x) = \frac{1}{1+e^{-x}}$

If the node should only activate when weighted sum $> 10 \Rightarrow$ Add a Bias

Learning $\rightarrow$ Finding the right weights and biases

Sigmoid $\rightarrow$ slow learner

ReLU (a) = $\max(0, a)$

$\hookrightarrow$ Rectified linear unit

$\Downarrow$
Biologically inspired

$\hookrightarrow$ either active or inactive



$$a_0^{(1)} = \sigma(w_{0,0} a_0^{(0)} + w_{0,1} a_1^{(0)} + \dots + w_{0,n} a_n^{(0)} + b_0)$$

$$\Downarrow$$

$$\sigma \left(\begin{bmatrix} w_{0,0} & w_{0,1} & \dots & w_{0,n} \\ w_{1,0} & w_{1,1} & \dots & w_{1,n} \\ \vdots & \vdots & \ddots & \vdots \\ w_{k,0} & w_{k,1} & \dots & w_{k,n} \end{bmatrix} \begin{bmatrix} a_0^{(0)} \\ a_1^{(0)} \\ \vdots \\ a_n^{(0)} \end{bmatrix} + \begin{bmatrix} b_0 \\ b_1 \\ \vdots \\ b_n \end{bmatrix} \right) \Rightarrow a^{(1)} = \sigma(Wa^{(0)} + b)$$

Neural Network from Scratch

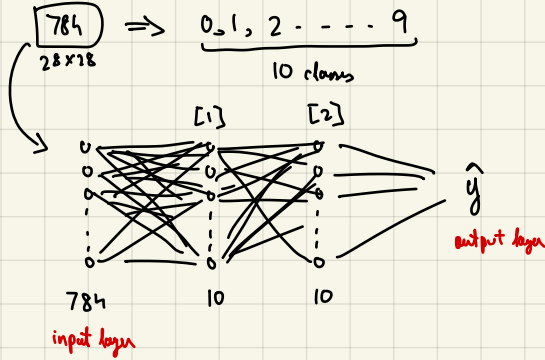
MNIST Database $\rightarrow 28 \times 28$ pixel handwritten digits

784
28 x 28

m training images

$$X = \begin{bmatrix} \text{784 column} \rightarrow 1 \text{ col} = 1 \text{ pixel} \\ x^{(1)} \\ x^{(2)} \\ \vdots \\ x^{(m)} \end{bmatrix}^T = \begin{bmatrix} | & | & \dots & | \\ x^{(1)} & x^{(2)} & \dots & x^{(m)} \\ | & | & \dots & | \end{bmatrix}$$

Pixel Value $\in [0, 255]$
B W



$$A^{[0]} = X \quad (784 \times m)$$

$$Z^{[1]} = W^{[1]} A^{[0]} + b^{[1]} \quad (10 \times 784 \quad 784 \times m \quad 10 \times 1 \rightarrow 10 \times m)$$

Activation function $\Rightarrow \text{ReLU}(x) = \begin{cases} x & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$

$$A^{[1]} = g(Z^{[1]}) = \text{ReLU}(Z^{[1]})$$

$$Z^{[2]} = W^{[2]} A^{[1]} + b^{[2]} \quad (10 \times m \quad 10 \times 10 \quad 10 \times m \quad 10 \times 1 \rightarrow 10 \times m)$$

$$A^{[2]} = \text{softmax}(Z^{[2]})$$

output layer

$$\begin{bmatrix} 1.3 \\ 5.1 \\ 2.2 \\ 0.7 \\ 1.1 \end{bmatrix} \rightarrow \begin{bmatrix} e^{z_i} \\ \sum_{j=1}^5 e^{z_j} \end{bmatrix} \rightarrow \begin{bmatrix} 0.02 \\ 0.9 \\ 0.05 \\ 0.01 \\ 0.02 \end{bmatrix}$$

start with prediction and measure the difference with the actual label

$$dZ^{[2]} = A^{[2]} - Y \quad (10 \times m \quad 10 \times m \quad 10 \times m)$$

$$dW^{[2]} = \frac{1}{m} dZ^{[1]} A^{[2]T} \quad (10 \times 10 \quad 10 \times m \quad m \times 10)$$

$$db^{[2]} = \frac{1}{m} \sum dZ^{[2]} \quad (10 \times 1 \quad 10 \times 1)$$

$$dZ^{[1]} = W^{[2]T} dZ^{[2]} \cdot g'(Z^{[1]}) \quad (10 \times m \quad 10 \times 10 \quad 10 \times m \quad 10 \times m)$$

$$dW^{[1]} = \frac{1}{m} dZ^{[1]} X^T \quad (10 \times 784 \quad 10 \times m \quad m \times 784)$$

$$db^{[1]} = \frac{1}{m} \sum dZ^{[1]} \quad (10 \times 1 \quad 10 \times 1)$$

backwards propagation

forward propagation

$$W^{[1]} = W^{[1]} - \alpha dW^{[1]}$$

$$b^{[1]} = b^{[1]} - \alpha db^{[1]}$$

$$W^{[2]} = W^{[2]} - \alpha dW^{[2]}$$

$$b^{[2]} = b^{[2]} - \alpha db^{[2]}$$

update parameters

$\alpha \rightarrow$ learning rate

$\hookrightarrow$ not trained by model